

TRN003 – Cloud Architecture on AWS

1. Course Overview

- **Duration:** 4 Days | 4 Hours per Day | Total 16 Hours
- **Delivery Mode:** Hybrid Training Model
- **Category:** Technical | Cloud Computing | DevOps | Architecture
- **Platform Focus:** Amazon Web Services (AWS)
- **Target Audience:** Cloud Engineers, DevOps Engineers, Infrastructure Teams, Solution Architects, Software Engineers

2. Course Objective

This program equips participants with enterprise-level AWS cloud architecture knowledge required to design scalable, secure, highly available, and cost-optimized cloud solutions using AWS services and modern cloud-native practices.

3. Learning Outcomes

- Understand core AWS architecture principles and cloud fundamentals.
- Design scalable and highly available cloud infrastructure on AWS.
- Build secure, resilient, and cost-optimized architectures.
- Deploy and manage compute, storage, networking, and database services.
- Implement monitoring, automation, and DevOps practices in AWS.
- Design end-to-end enterprise cloud solutions using architectural best practices.

4. Curriculum Structure

DAY 1 – AWS CLOUD FOUNDATIONS

Session	Session 1: Introduction to Cloud Computing
Duration	60 Minutes
Topics Covered	Cloud Computing Models, AWS Global Infrastructure, Regions & Availability Zones, Shared Responsibility Model
Lab / Workshop	AWS Environment Overview
Expected Outcome	Understand AWS ecosystem and architecture fundamentals.

Session	Session 2: AWS Core Services
Duration	60 Minutes
Topics Covered	EC2, S3, EBS, IAM, VPC Basics, RDS Overview
Lab / Workshop	Deploy a Basic EC2 Instance
Expected Outcome	Hands-on exposure to foundational AWS services.

Session	Session 3: AWS Networking Fundamentals
Duration	60 Minutes
Topics Covered	VPC Architecture, Subnets, Route Tables, Internet Gateway, NAT Gateway, Security Groups, NACLs
Lab / Workshop	Build Secure VPC Architecture
Expected Outcome	Design secure and isolated cloud networks.

Session	Session 4: Identity & Access Management
Duration	60 Minutes
Topics Covered	IAM Users, Roles, Policies, MFA, Security Best Practices
Lab / Workshop	Role-Based Access Configuration
Expected Outcome	Implement cloud security and access management.

DAY 2 – DESIGNING SCALABLE CLOUD INFRASTRUCTURE

Session	Session 5: Compute & Scaling Services
Duration	60 Minutes
Topics Covered	Auto Scaling Groups, Load Balancers, Launch Templates, Elastic Beanstalk
Lab / Workshop	Configure Auto Scaling
Expected Outcome	Build scalable and highly available applications.

Session	Session 6: Storage & Database Services
Duration	60 Minutes
Topics Covered	S3 Storage Classes, EFS, Glacier, RDS, DynamoDB
Lab / Workshop	Enterprise Storage Architecture
Expected Outcome	Select appropriate storage and database solutions.

Session	Session 7: High Availability & Disaster Recovery
Duration	60 Minutes
Topics Covered	Backup Strategies, Multi-AZ Deployments, Disaster Recovery Models, Failover Concepts
Lab / Workshop	DR Architecture Case Study
Expected Outcome	Design resilient enterprise systems.

Session	Session 8: Cost Optimization & Monitoring
Duration	60 Minutes
Topics Covered	CloudWatch, AWS Budgets, Cost Explorer, Trusted Advisor
Lab / Workshop	AWS Cost Optimization Workshop
Expected Outcome	Optimize cloud resources and operational cost.

DAY 3 – DEVOPS & CLOUD AUTOMATION

Session	Session 9: Infrastructure as Code
Duration	60 Minutes
Topics Covered	CloudFormation Basics, Infrastructure Automation, Stack Management
Lab / Workshop	Deploy Infrastructure Template
Expected Outcome	Automate repeatable cloud deployments.

Session	Session 10: CI/CD on AWS
Duration	60 Minutes
Topics Covered	CodeCommit, CodeBuild, CodeDeploy, CodePipeline
Lab / Workshop	CI/CD Pipeline Demonstration
Expected Outcome	Implement DevOps practices on AWS.

Session	Session 11: Containers & Serverless
Duration	60 Minutes
Topics Covered	Docker Basics, ECS, EKS Overview, Lambda, API Gateway
Lab / Workshop	Serverless Application Deployment
Expected Outcome	Understand modern cloud-native architectures.

Session	Session 12: Security & Compliance
Duration	60 Minutes
Topics Covered	AWS Security Services, Encryption, Compliance Frameworks, Logging & Auditing
Lab / Workshop	Security Best Practices Exercise
Expected Outcome	Implement enterprise-grade cloud security.

DAY 4 – ENTERPRISE CLOUD ARCHITECTURE

Session	Session 13: Designing Enterprise Solutions
Duration	60 Minutes
Topics Covered	Architecture Frameworks, Well-Architected Framework, Multi-Tier Applications
Lab / Workshop	Enterprise Architecture Discussion
Expected Outcome	Design real-world enterprise architectures.

Session	Session 14: Migration Strategies
Duration	60 Minutes
Topics Covered	Cloud Migration Approaches, Lift-and-Shift, Replatforming, Hybrid Cloud
Lab / Workshop	Migration Assessment Workshop
Expected Outcome	Plan cloud migration strategies.

Session	Session 15: Real-World Case Studies
Duration	60 Minutes
Topics Covered	E-Commerce, Banking, Healthcare, SaaS Cloud Architectures
Lab / Workshop	Architecture Review Exercise
Expected Outcome	Analyze industry cloud implementation patterns.

Session	Session 16: Final Capstone Project
Duration	60 Minutes
Topics Covered	Enterprise Solution Design & Architecture Presentation
Lab / Workshop	AWS Architecture Capstone
Expected Outcome	Present end-to-end cloud solution designs.

5. Hands-On Labs & Activities

- AWS Console Navigation & Environment Setup
- Create and Configure EC2 Instances
- Build Secure Virtual Private Cloud (VPC)
- Implement Auto Scaling Architecture
- Configure CI/CD Pipeline on AWS
- Deploy Serverless Applications using Lambda
- Cloud Cost Optimization Exercise
- Enterprise Cloud Solution Architecture Workshop

6. Assessment Framework

Assessment Component	Weightage
Hands-On Labs	30%
Knowledge Checks & Quizzes	20%
Architecture Workshops	20%
Final Capstone Project	30%

7. Final Capstone Project

Participants design and present an enterprise AWS cloud architecture solution for a business use case.

- Design scalable infrastructure architecture.
- Implement networking and security strategy.
- Select suitable storage and database services.
- Define DevOps automation and deployment approach.
- Present architecture diagrams and business justification.

8. Expected Deliverables

- AWS Architecture Design Document
- Cloud Infrastructure Templates
- Cloud Security & Deployment Plan
- Cloud Cost Optimization Report
- Enterprise Cloud Architecture Presentation

9. Expected Business Outcomes

- Accelerate enterprise cloud adoption initiatives.
- Improve infrastructure scalability and availability.
- Reduce operational overhead through automation.
- Enhance cloud security and governance practices.
- Strengthen DevOps and cloud engineering capabilities.
- Enable cost-efficient and resilient cloud architectures.

10. Training Methodology

Concepts & Architecture Theory: 40% | Hands-On Labs & Workshops: 50% | Case Studies & Discussions: 10%